MLOps Assignment 2

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Experiment: sentiment-inference-lab (MLflow)

Results Summary

Run	batch_size	accuracy	avg_confidence	total_latency_sec
run_batch1	1	1.0	~1.00	~0.34
run_batch4	4	1.0	0.99965	0.15693
run_batch8	8	1.0	0.99965	0.18375

1. Which run had the lowest latency, and why might that be?

run_batch4 had the lowest latency at about 0.157 seconds. run_batch8 followed at about 0.184 seconds, and run_batch1 was slowest at about 0.34 seconds.

Batching lets the model handle several sentences in one forward pass, so the fixed overhead is shared instead of being paid for every sentence. That explains why batch sizes 4 and 8 both beat batch size 1. The very first run also carries model warm-up cost, which adds to its time. Batch size 8 was slightly slower than 4 here; with only 10 sentences, a batch of 8 leaves a small second batch and adds padding work, so the extra gain from a bigger batch is limited.

2. Did accuracy change between runs? Why or why not?

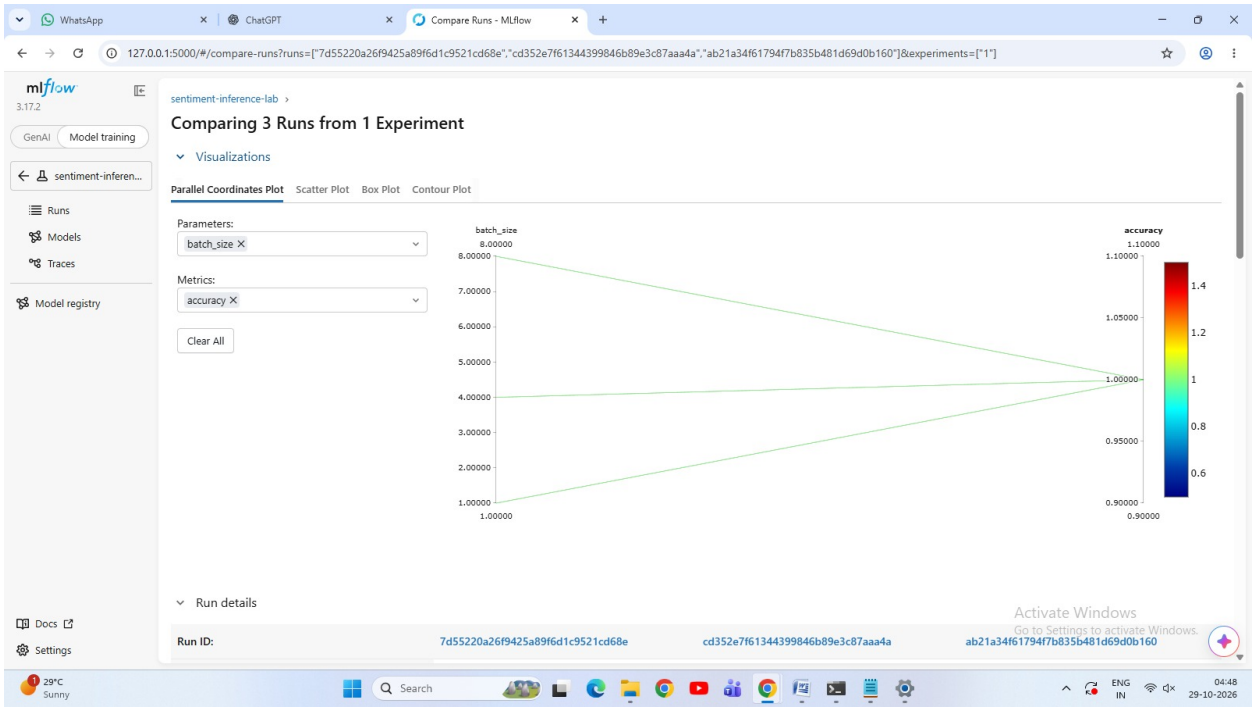
No. Accuracy stayed at 1.0 (100%) in all three runs. Every run used the same pre-trained model (distilbert-base-uncased-finetuned-sst-2-english) on the same 10 labelled sentences. Batch size only changes how the inputs are grouped for processing, not what the model predicts, so the predicted labels and the accuracy stayed the same.

3. Name two benefits of using MLflow here instead of printing results in the terminal.

(a) Record-keeping and reproducibility: MLflow stores the parameters, metrics and prediction files of every run, so the results can be reopened and checked later instead of vanishing when the terminal closes.

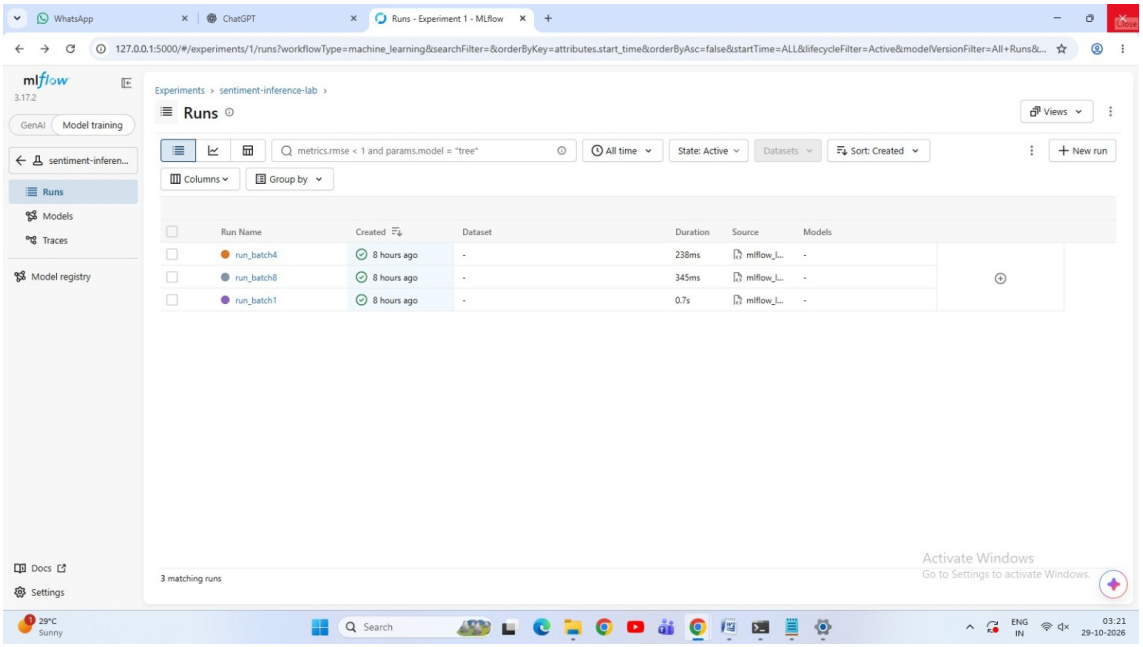
(b) Easy comparison: The MLflow interface lets us compare runs side by side through tables and plots (for example the parallel coordinates plot), which makes differences in latency or accuracy easier to see than scrolling through printed text.

Screenshot of the Compare view:

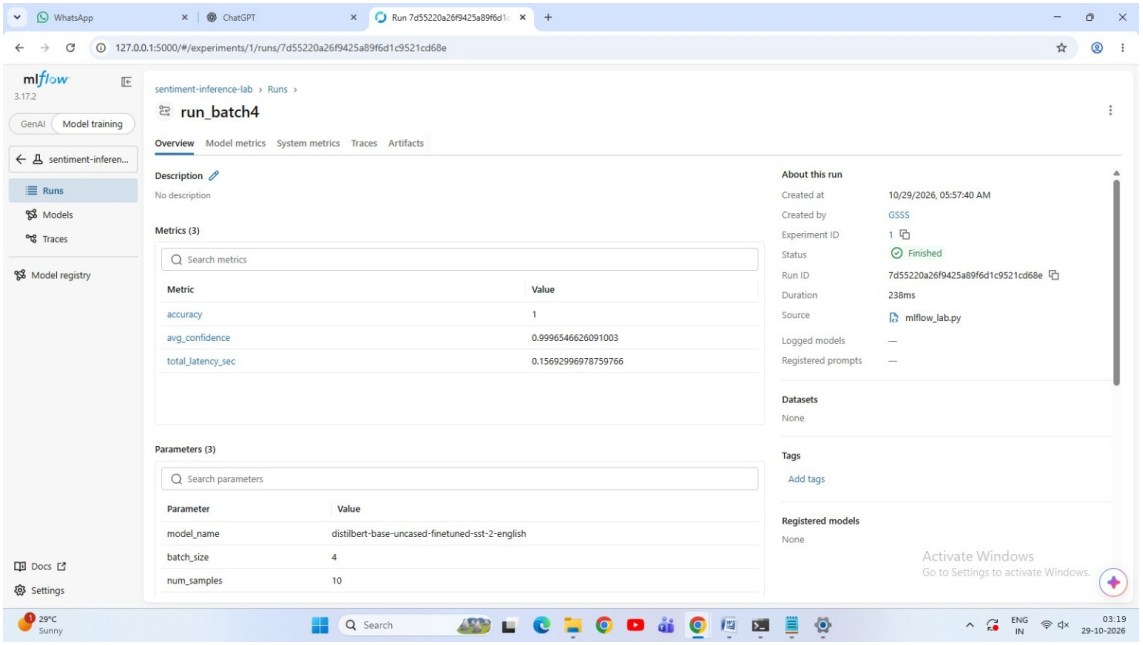


Parallel coordinates plot comparing the three runs (batch_size 1, 4 and 8) against accuracy. All three lines end at accuracy 1.0.

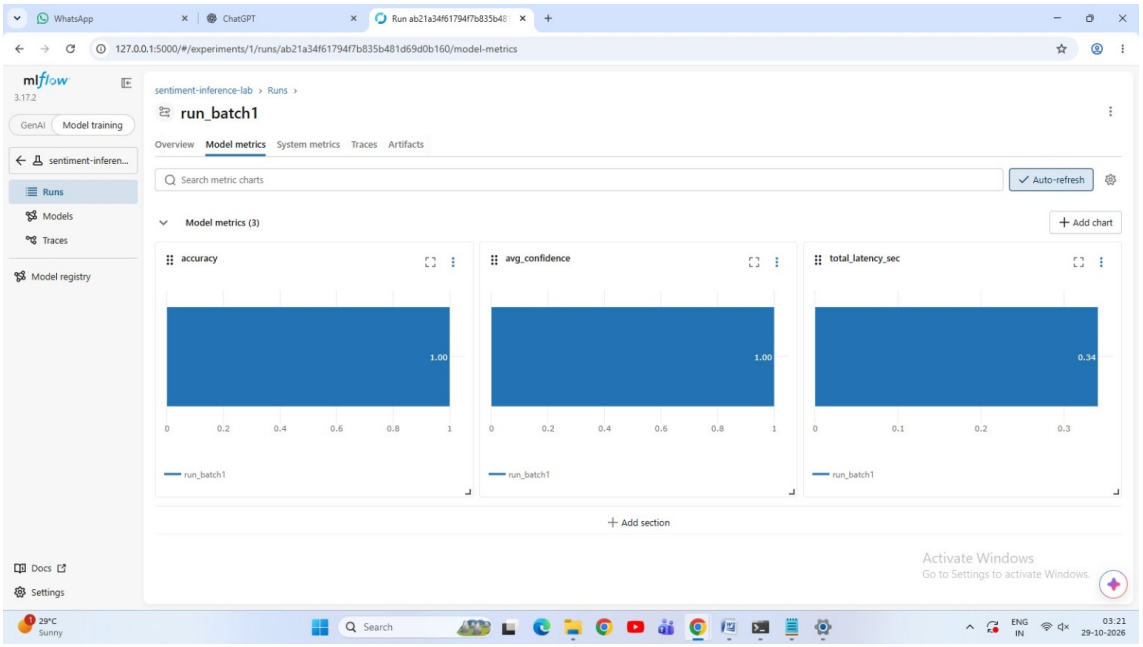
Screenshot of the MLflow runs table:



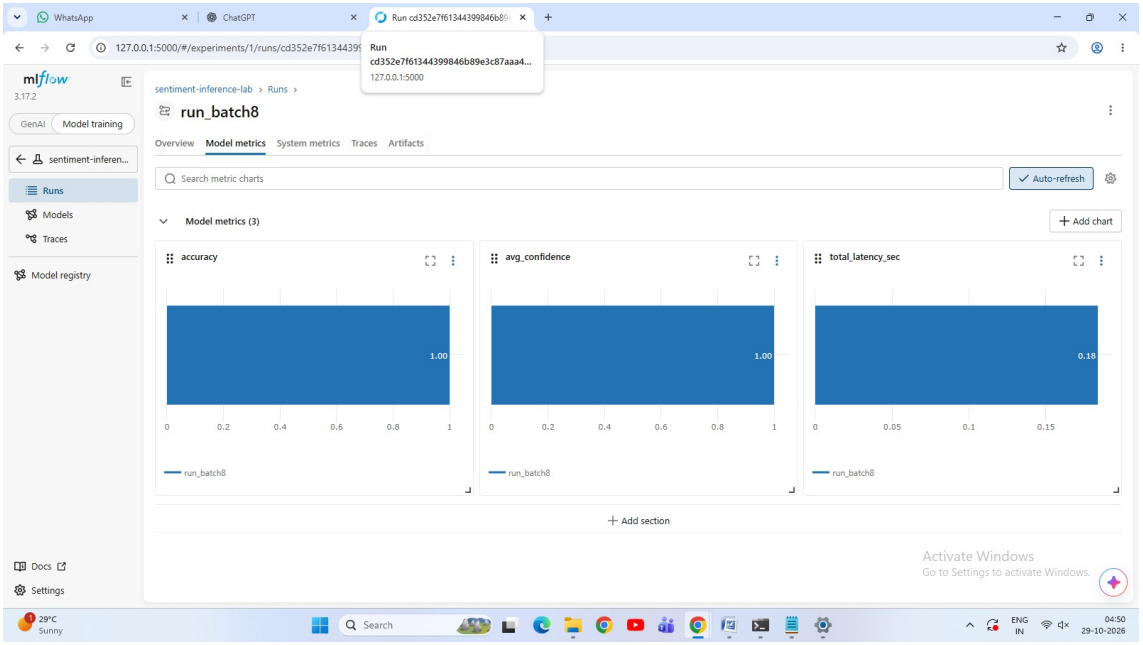
Overview of run_batch4 (batch_size 4):



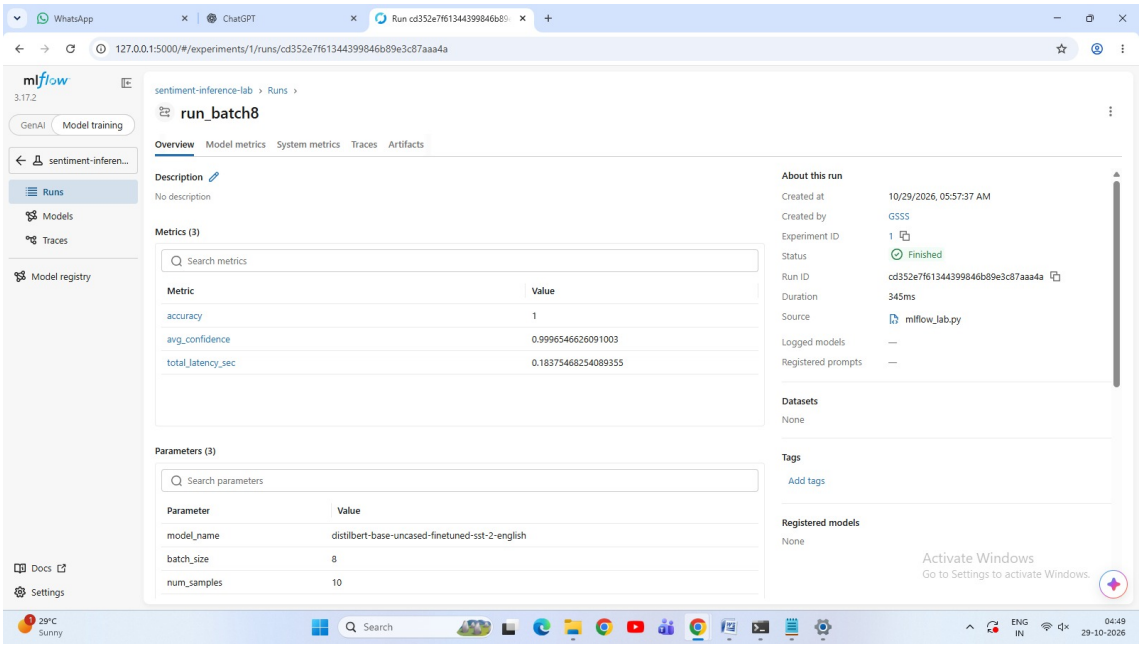
Model metrics of run_batch1:



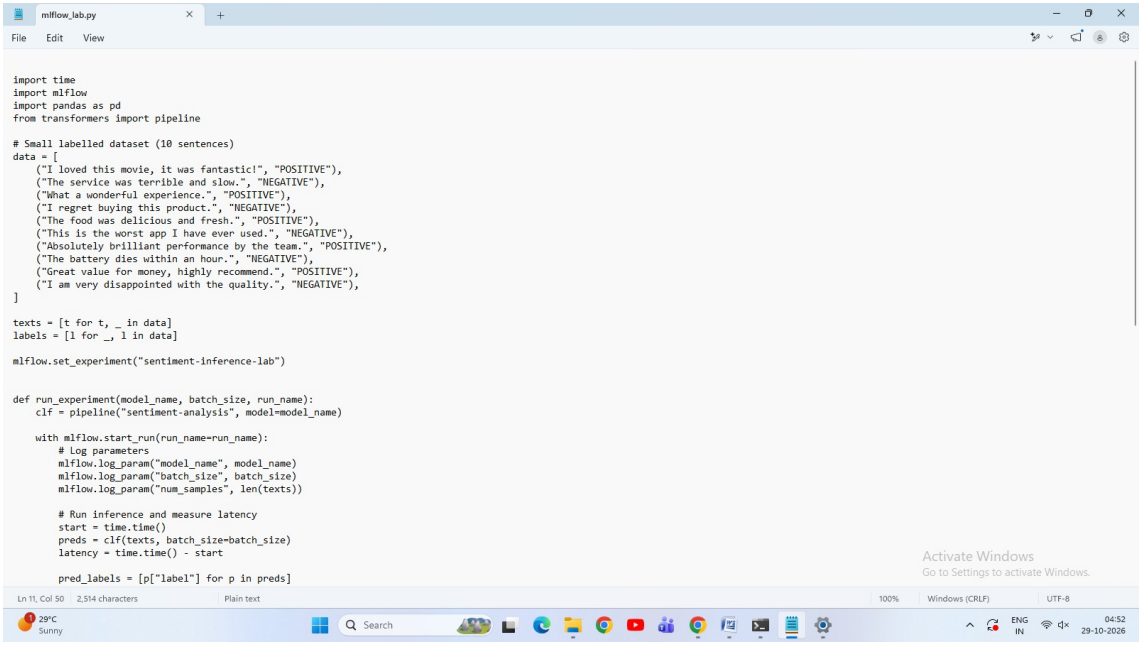
Model metrics of run_batch8:



Overview of run_batch8 (batch_size 8):



mlflow_lab.py file:



Execution (environment setup and package installation):

